C2 Lakehouse, Warehouse and SQL analytics endpoint

When to use each.

C2-L1 Lakehouse Tables and Files, and schemas

Raw files in Files, Delta tables in Tables, and schema-qualified names in shared code.

Lakehouse Tables vs Files	Tables: managed Delta tables, visible to SQL and Direct Lake. Files: raw and non-Delta data.
Non-Delta folder under Tables	Shown in the Unidentified area, not queryable.
Lakehouse schemas	On by default for new lakehouses. dbo is the fixed default schema.
Spark four-part name	workspace.lakehouse.schema.table. A bare name means dbo in the default lakehouse.

C2-L2 Warehouse T-SQL, and where it differs from SQL Server

Map the types, declare keys as NOT ENFORCED hints, and check uniqueness yourself after every load.

Primary keys in Fabric Warehouse	Only PRIMARY KEY NONCLUSTERED ... NOT ENFORCED. Duplicates are not blocked.
nvarchar, money, datetime in Fabric Warehouse	Not supported. Use varchar, decimal, datetime2.
MERGE and TRUNCATE in Fabric Warehouse	Both supported. MERGE is generally available.
Triggers in Fabric Warehouse	Not supported. Put the logic in the loading procedure.

C2-L3 Lakehouse, Warehouse or SQL analytics endpoint

The data and the logic are the same. The item type decides whether the procedure runs at all, and whether a failure leaves the tables consistent.

Lakehouse, Warehouse or endpoint: the deciding question	Who writes the data, and in which language? Spark writes to a lakehouse, T-SQL writes to a warehouse.
What can you do on the SQL analytics endpoint?	SELECT and create views, functions and procedures. No INSERT, UPDATE or DELETE on table data.
When do you need a Warehouse over a Lakehouse?	T-SQL writes, stored procedure loads, and multi-table transactions.
Read a lakehouse table from a warehouse in the same workspace	Use a three-part name: lakehouse.schema.table. No copy.
A new lakehouse table is missing from the endpoint	Metadata sync lag. Usually under a minute. Refresh or retry before dependent steps.